MEDICATION SAFETY ASSISTANT

Database Design Report

(PA2 – Analysis and Design)

Version 1.0

Document identifier: PA2-Group02-DatabaseDesign
Prepared by: Group 02
Organization: University of Science, VNU-HCM
Date: 29 October 2026

Revision History

Date	Version	Description	Author
29 Oct 2026	1.0	Initial PA2 database design report derived from the approved ERD views and current CSV data profile	Group 02

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1. Introduction

1.1 Purpose

This report defines the database design for the Medication Safety Assistant. It translates the current conceptual and SQL-style ERD views into a coherent logical data model, explains how data moves through the system, and records the constraints that must be preserved when the model is converted into a physical database schema.

The report is intended to support four activities:

- implementation of the relational database and migration scripts;
- review of entity ownership and relationships across the five product modules;
- controlled ingestion and medical review of the six current CSV datasets; and
- verification that clinical, operational, and support workflows do not receive broader access than required.

1.2 Scope

The design covers 64 logical entities grouped into the following areas:

- user identity, role-based access control, professional profiles, and audit logging;
- medication intake, document extraction, confirmation, and prescription records;
- versioned medication, ingredient, indication, safety, and evidence catalogs;
- medication identification and deterministic drug–drug interaction assessment;
- HCP review and direct patient–HCP communication;
- medication regimens, reminders, adherence, notifications, and technical support; and
- import, validation, medical-data review, release, and traceability of reference data.

This is a logical database design report. It identifies keys and principal relationships but does not yet prescribe every column, SQL data type, check constraint, partition, or database-engine-specific object. Those items belong to the physical design phase.

1.3 Source Artifacts

The report is based on the following repository artifacts:

- `docs/analysisanddesign/Unified-SQL-ERD.md`;
- `docs/analysisanddesign/Module-ERD-Views.md`;
- `docs/analysisanddesign/Shared-Domain-ERD-Views.md`;
- the eleven PNG diagrams generated from those Mermaid definitions; and
- `data/Data-Overview.md` and the six profiled CSV datasets.

1.4 Audience

The primary audience includes backend developers, database designers, testers, the product owner, HCP representatives, and medical data reviewers. Developers use the report to implement the schema; reviewers use it to verify that evidence and approval states are not bypassed; testers use it to derive integrity, authorization, and data-lifecycle test cases.

1.5 Terminology

Table 1: Key database-design terminology

Term	Meaning in this report
Candidate	Imported or inferred data that has not yet been accepted for operational clinical use.
Operative	Data that may participate in identification or safety assessment after required review and release.
Dataset version	An immutable workspace or snapshot that groups related reference-data rows produced by one import/review cycle.
Dataset release	A controlled publication of one approved dataset version for use by runtime modules.
HCP	A healthcare professional who may care for patients; this role does not automatically grant medical-data-review authority.
Medical reviewer	A qualified user explicitly assigned to review medication, mapping, interaction, or evidence data within an approved scope.
Safety drug entity	A normalized drug identifier in a safety source namespace, such as DDInter, used as an interaction endpoint.
Snapshot	The patient's medication list frozen at an assessment point so that results remain reproducible.

2. Data and Design Basis

2.1 Current CSV Dataset

The current data package contains 42,903 rows in six transformed CSV files. These files are neither raw source documents nor a fully approved production catalog. They must therefore enter the database through a versioned import and review process.

Table 2: Current CSV inventory and intended logical targets

File	Rows	Columns	Primary logical targets
`medications.csv`	10,670	30	`MEDICATIONS`; registration, manufacturer, dosage form, route, packaging, and verification state.
`ingredients.csv`	1,940	16	`INGREDIENTS`; local and accepted ingredient identities and verification scope.
`medication_ingredients.csv`	12,899	25	`MEDICATION_COMPONENTS` and `STRENGTH_ASSERTIONS`; source text, accepted ingredient link, chemical form, and strength parsing state.
`medication_indications.csv`	5,614	16	`MEDICATION_INDICATIONS`; candidate indication statements pending medical normalization and approval.
`medication_priorities.csv`	6,105	15	`MEDICATION_PRIORITIES`; collection and review priority, not clinical prescribing priority.

File	Rows	Columns	Primary logical targets
ddinter_mappings.csv	5,675	17	SAFETY_DRUG_ENTITIES and COMPONENT_SAFETY_MAPPINGS; candidate component-to-DDInter mappings.
Total	**42,903**	—	Versioned reference-data import and review workflow.

2.2 Important Data-Quality Findings

The logical design preserves incomplete and disputed source information rather than deleting it:

- 8,741 of 10,670 medication candidates are Level-2 qualified, while 1,929 are not.
- 9,313 medications have no explicitly supported route; the system must not infer oral administration.
- 1,540 ingredient identities are auto-verified source-local labels, while 400 remain source-asserted only.
- 2,657 medication components have no accepted ingredient identifier, including 662 intentionally unresolved components.
- all 5,614 indication rows are candidates; no row is currently an approved product indication.
- all 5,675 DDInter mappings are candidates with ACCEPTED=FALSE and OPERATIVE=FALSE.

Consequently, null foreign keys and review states are meaningful business states, not merely data defects. The physical design must distinguish "unknown or awaiting review" from "not applicable" and from "rejected."

2.3 Data Not Present in the Current CSV Package

The current six files do not contain interaction pairs, interaction severity, or evidence for each pair. Therefore, they cannot directly populate DRUG_INTERACTIONS, DRUG_INTERACTION_PARTICIPANTS, or INTERACTION_EVIDENCE. Module 3 currently expects a separate DDInter interaction file containing DDInterID_A, Drug_A, DDInterID_B, Drug_B, and Level.

The absence of a pair from a loaded interaction dataset must never be interpreted as proof that no interaction exists. Runtime results must retain the dataset release and evidence context used for the assessment.

2.4 Design Principles

1. **Trace every decision to evidence.** Source records, hashes, validation results, reviewer decisions, and evidence links are retained.
2. **Separate candidate from operative data.** Imported or machine-generated mappings cannot affect patient-facing safety results until approved and released.
3. **Make assessments reproducible.** A safety assessment references an immutable medication-list snapshot and an explicit dataset release.
4. **Separate clinical care from data governance.** HCP care privileges and medical-data-review privileges are independently assigned.
5. **Apply least privilege.** Technical support handles tickets and diagnostic context, not clinical conclusions; patient access is constrained to owned records; HCP access is constrained by care relationships.
6. **Prefer deterministic safety logic.** Drug interaction matching uses reviewed identifiers and relational joins. Statistical similarity is not an authoritative safety decision.

3. Logical Database Architecture

3.1 Unified Model

The unified ERD contains 64 entities. It provides the authoritative cross-module view: user and access entities are shared by care, review, and support workflows; a released reference dataset supports medication identification and safety assessment; assessment results flow into HCP review and then into an approved regimen and reminder schedule.

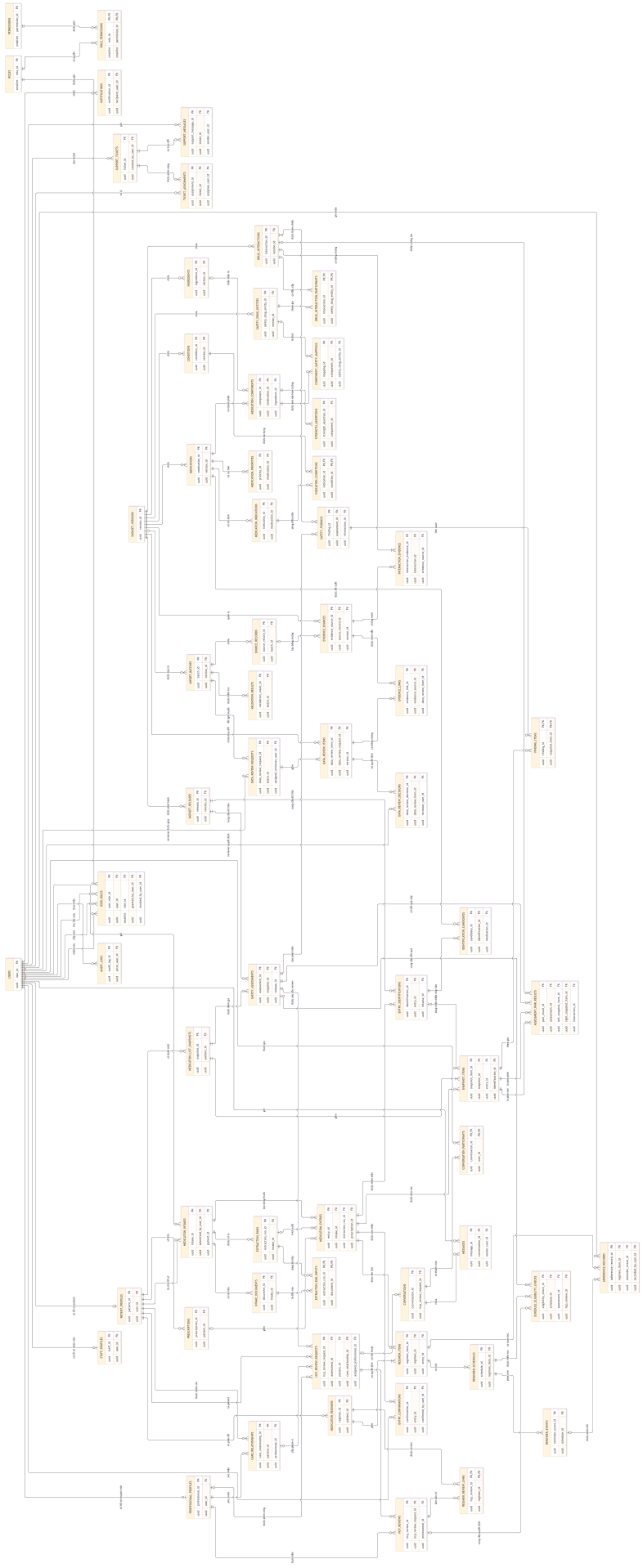

Figure 1: Unified SQL-style ERD for the Medication Safety Assistant

The unified view is intentionally dense. It is used for cross-domain integrity review and implementation planning, while Sections 4 and 5 provide smaller views for normal reading.

3.2 Entity Groups

Table 3: Logical entity groups

Group	Count	Responsibility
Identity, access, and profiles	10	Accounts, roles, permissions, patient/professional/staff profiles, care scope, and audit records.
Medication intake	7	Submission, document capture, extraction, prescription grouping, medication entries, and patient confirmation.
Data import and governance	11	Dataset version/release, batches, source records, validation, review requests/items/decisions, and evidence.
Medication reference and identification	10	Medication, ingredient, components, strength, indications, conditions, priorities, and entry identification.
Safety assessment	10	Safety identifiers and mappings, interactions, participants, evidence, medication snapshots, pair results, and findings.
HCP review and communication	8	Review requests/decisions, care conversation, participants, and messages.
Regimen, reminder, and adherence	8	Regimens, items, review links, schedules, eligibility, events, adherence, and notifications.
Technical support	3	Tickets, assignment history, and support messages.

Some entities participate in multiple groups; the counts above describe primary ownership and total 64 distinct tables in the unified model.

3.3 Key Architectural Boundaries

3.3.1 Identity versus Domain Profiles

`USERS` stores authentication identity. Domain-specific data is held in `PATIENT_PROFILES`, `PROFESSIONAL_PROFILES`, and `STAFF_PROFILES`. This prevents support staff and administrators from receiving unnecessary health-profile columns merely because they have user accounts.

3.3.2 Roles versus Qualifications

`USER_ROLES` grants application roles. A professional qualification belongs in `PROFESSIONAL_PROFILES`; medical reviewer authority is granted through a role and recorded assignment. An HCP can also be a medical reviewer, but HCP status alone does not authorize reference-data approval.

3.3.3 Operational Data versus Reference Data

Patient-generated data such as medication entries and snapshots remains separate from versioned reference data. `ENTRY_IDENTIFICATIONS` bridges the two. `SAFETY_ASSESSMENTS` records the release used so a previous result is not silently changed when a new reference catalog is published.

3.3.4 Clinical Communication versus Technical Support

`CONVERSATIONS` and `MESSAGES` support direct patient–HCP discussion within a review case. `SUPPORT_TICKETS` and `SUPPORT_MESSAGES` support login, upload, OCR, and application incidents. Technical support is not placed

between the patient and the HCP.

3.4 Cardinality Notation

The diagrams use Crow's Foot notation. A circle denotes an optional relationship (minimum zero), a bar denotes exactly one at that end, and the crow's foot denotes many. For example, USERS||--o\{SUPPORT_TICKETS means one user may create zero or many tickets, while every ticket has one creator. These symbols describe logical minimum and maximum participation; corresponding nullability, uniqueness, and foreign-key constraints must be implemented in the physical schema.

4. Module-Oriented Database Design

The following views show the tables each product module reads or writes. Shared entities may appear in more than one view because module ownership and relational ownership are not identical.

4.1 Module 1 – Medication Intake

Module 1 captures medication information from text, image, or document input. A user creates MEDICATION_INTAKES; each submission may contain INTAKE_DOCUMENTS. One or more EXTRACTION_RUNS consume those documents through EXTRACTION_RUN_INPUTS. Parsed rows become MEDICATION_ENTRIES, optionally grouped by PRESCRIPTIONS, and the patient records confirmation or correction in ENTRY_CONFIRMATIONS.

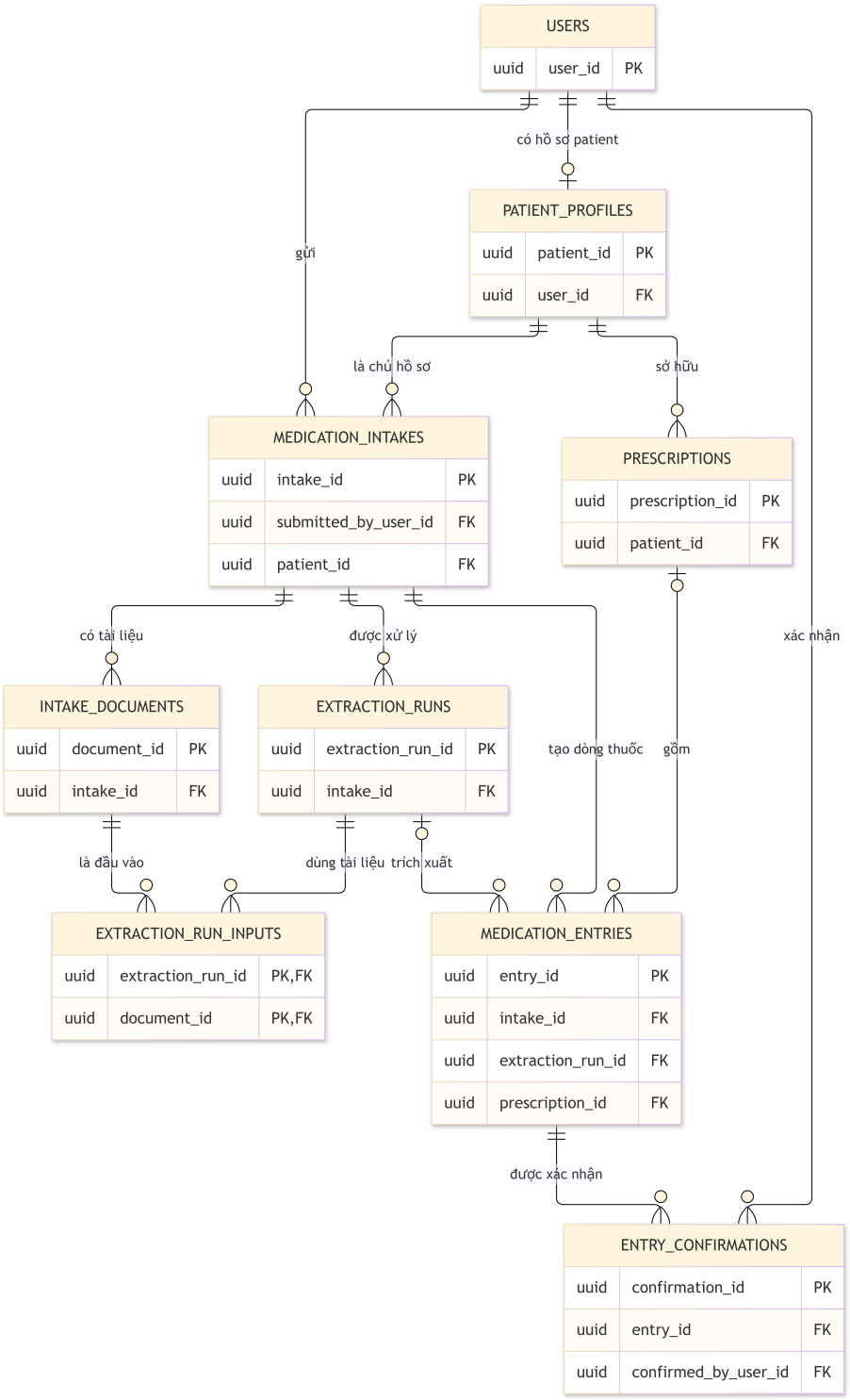

Figure 2: Module 1 – Medication Intake ERD

Table 4: Module 1 tables and responsibilities

Table	Responsibility
USERS	Identifies the submitter without duplicating account data in the intake domain.
MEDICATION_INTAKES	Represents one intake session and records input method, owner, status, and submission context.
INTAKE_DOCUMENTS	Stores metadata and secure object references for images or documents; binary content should reside in object storage.
EXTRACTION_RUNS	Records an OCR/parser execution, model or rule version, timestamps, and outcome.
EXTRACTION_RUN_INPUTS	Many-to-many bridge between extraction runs and the documents they process.
PRESCRIPTIONS	Groups medication entries that belong to the same patient prescription or source document.
MEDICATION_ENTRIES	Holds raw text, extracted fields, status, and the patient-facing medication row.
ENTRY_CONFIRMATIONS	Keeps the actor, decision, correction, and time of confirmation instead of overwriting extraction history.

4.2 Module 2 – Medication Identification and Reference

Module 2 resolves a confirmed medication entry against one released medication catalog. ENTRY_IDENTIFICATIONS records an identification attempt and selected result; IDENTIFICATION_CANDIDATES retains alternatives and scores. The selected MEDICATIONS row expands through components, ingredients, strengths, indications, and conditions.

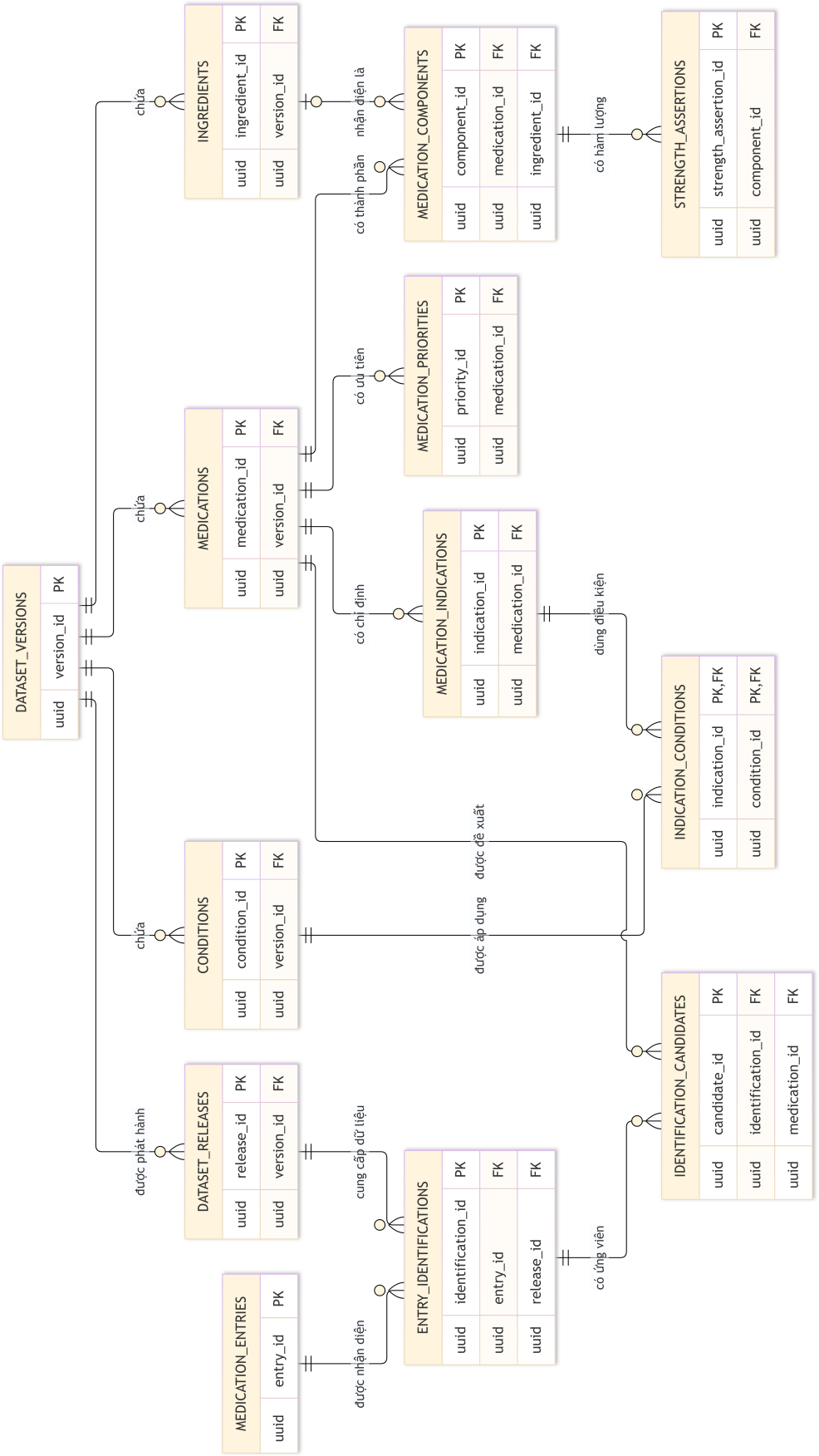

Figure 3: Module 2 – Medication Identification and Reference ERD

Table 5: Module 2 principal tables and responsibilities

Table	Responsibility
DATASET_RELEASES	Selects the approved reference-data release used by the identification attempt.
ENTRY_IDENTIFICATIONS	Records input entry, release, method, status, confidence, selected medication, and reviewer state.
IDENTIFICATION_ CANDIDATES	Preserves ranked candidates so ambiguity can be reviewed and evaluated.
MEDICATIONS	Versioned product identity, including name, registration, manufacturer, dosage form, route, and verification status.
INGREDIENTS	Versioned ingredient identity with source-local and accepted normalized names.
MEDICATION_COMPONENTS	Connects a medication to a component and, when accepted, to an ingredient identity.
STRENGTH_ASSERTIONS	Stores raw and normalized strength assertions separately from component identity.
MEDICATION_INDICATIONS	Stores source indication statements and approval state.
CONDITIONS	Stores reviewed condition concepts.
INDICATION_CONDITIONS	Many-to-many link between approved indication statements and normalized conditions.
MEDICATION_PRIORITIES	Stores collection/review priority; it must not be interpreted as clinical recommendation.

4.3 Module 3 – Medication Safety Assessment

Module 3 freezes the patient's current medication list in MEDICATION_LIST_SNAPSHOTS. SNAPSHOT_ITEMS reference the exact entries and identifications included. A SAFETY_ASSESSMENTS row binds that snapshot to a dataset release. Pairwise results are retained in ASSESSMENT_PAIR_RESULTS; user-facing findings are stored in SAFETY_FINDINGS and linked to affected items through FINDING_ITEMS.

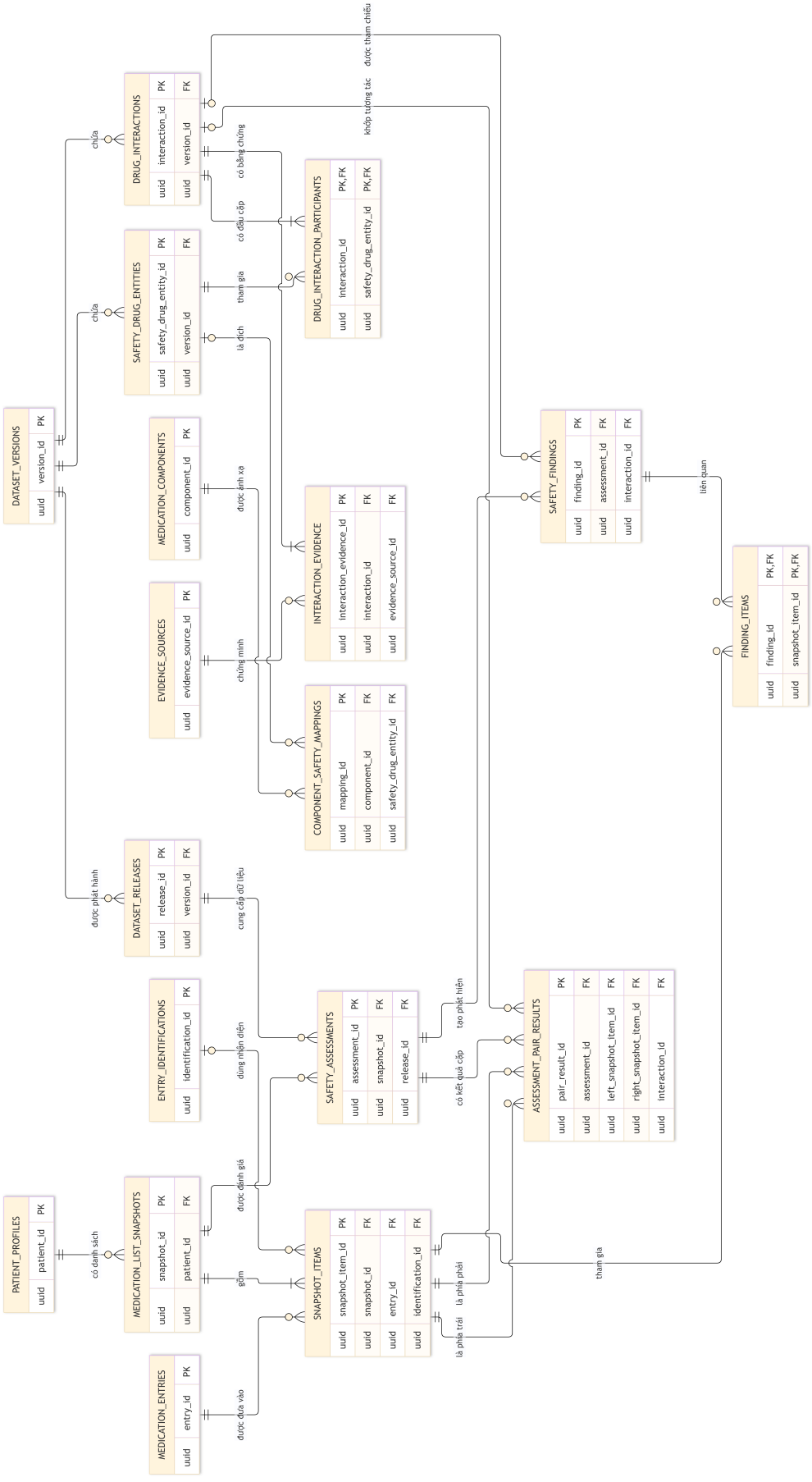

Figure 4: Module 3 – Medication Safety Assessment ERD

Table 6: Module 3 principal tables and responsibilities

Table	Responsibility
MEDICATION_LIST_SNAPSHOTS	Immutable assessment-time list owned by a patient.
SNAPSHOT_ITEMS	Connects snapshot rows to medication entries and accepted identification results.
SAFETY_ASSESSMENTS	Assessment execution, dataset release, lifecycle status, and timestamps.
ASSESSMENT_PAIR_RESULTS	Canonical unordered pair evaluation with matched interaction and outcome.
SAFETY_FINDINGS	Patient/HCP-facing finding with severity, explanation, and referenced interaction.
FINDING_ITEMS	Links one finding to the medication items involved.
SAFETY_DRUG_ENTITIES	Reviewed endpoint identifiers from the interaction namespace.
COMPONENT_SAFETY_MAPPINGS	Reviewed component-to-safety-entity mapping; candidate mappings are non-operative.
DRUG_INTERACTIONS	Canonical interaction record independent of source-specific participant order.
DRUG_INTERACTION_PARTICIPANTS	Interaction endpoints; the physical model must enforce exactly two distinct participants per pair.
INTERACTION_EVIDENCE	Connects interactions to one or more evidence sources and preserves disagreement.

4.4 Module 4 – HCP Review

Module 4 creates an explicit request for professional interpretation of a safety assessment. The request records patient, assessment, care relationship, assignee, priority, and status. HCP_REVIEWS retains the professional decision. A linked conversation permits the patient and assigned HCP to communicate directly.

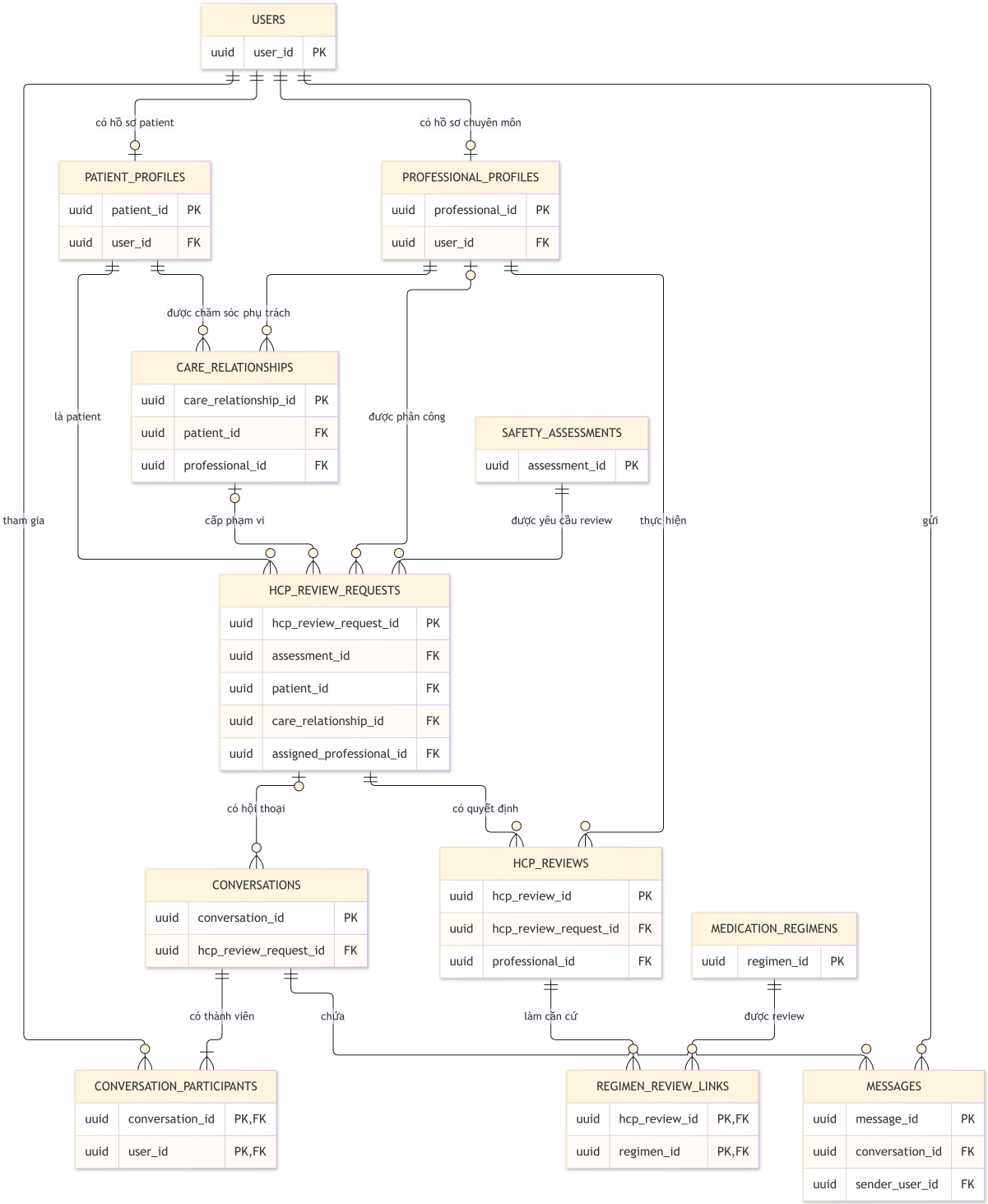

Figure 5: Module 4 – HCP Review ERD

Table 7: Module 4 tables and responsibilities

Table	Responsibility
CARE_RELATIONSHIPS	Establishes the professional's authorized patient scope.
HCP_REVIEW_REQUESTS	Case assignment and workflow state for review of one assessment.
HCP_REVIEWS	Professional interpretation, decision, recommendations, and completion metadata.
CONVERSATIONS	Case-bound communication channel, not a general public chat.
CONVERSATION_ PARTICIPANTS	Explicit participant membership and access boundary.
MESSAGES	Immutable message content and sender metadata within the authorized conversation.

4.5 Module 5 – Medication Reminders

Module 5 schedules reminders only from a regimen associated with patient entries and, where required, HCP review. Eligibility checks prevent an unconfirmed or clinically blocked item from producing reminder events. Adherence is recorded separately from delivery events.

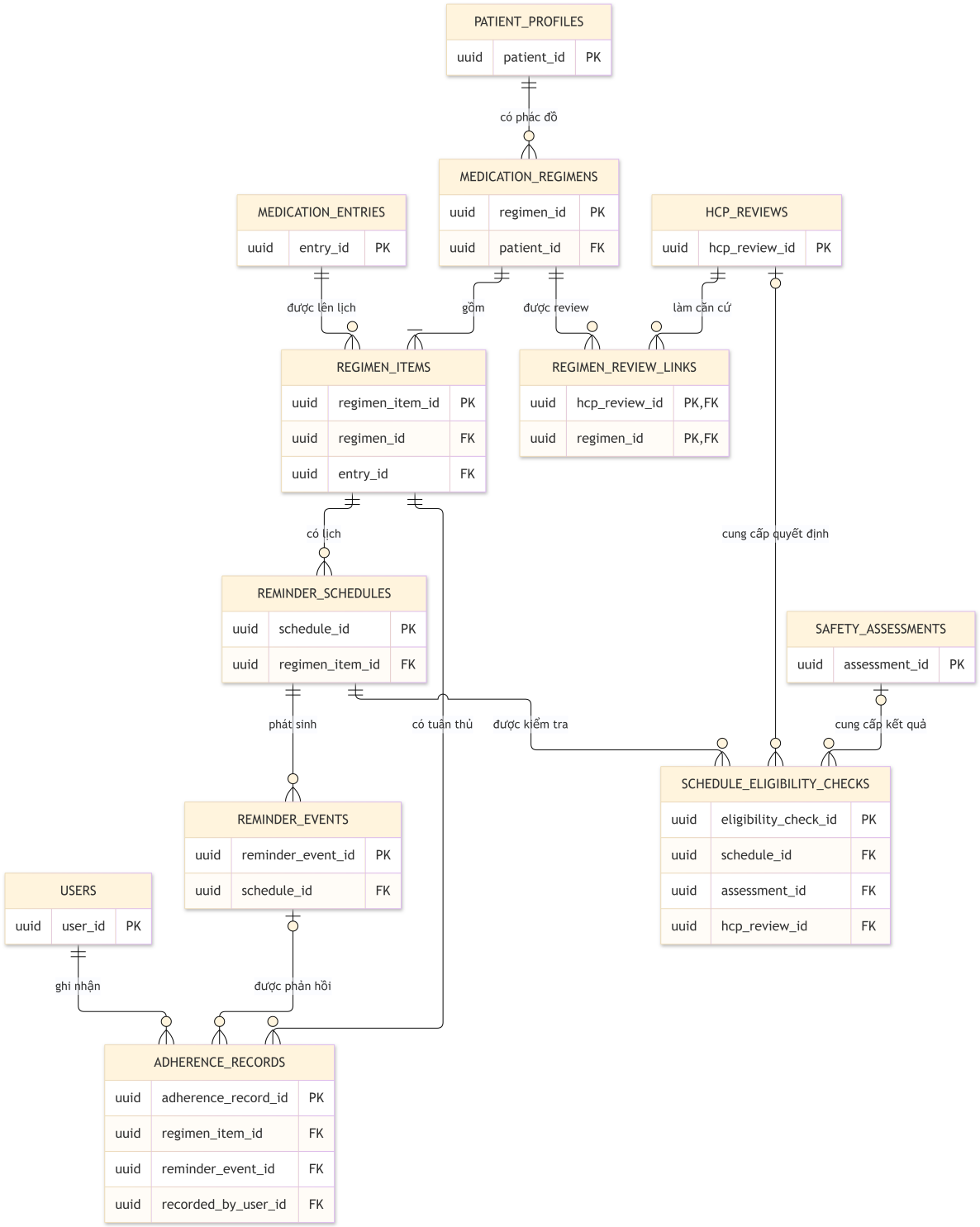

Figure 6: Module 5 – Medication Reminders ERD

Table 8: Module 5 tables and responsibilities

Table	Responsibility
MEDICATION_REGIMENS	Patient regimen header and lifecycle state.
REGIMEN_ITEMS	Instructions and date range for one medication entry within a regimen.
REGIMEN_REVIEW_LINKS	Records which HCP review supports a regimen.
REMINDER_SCHEDULES	Frequency, timing, timezone, start/end, and schedule status.
SCHEDULE_ELIGIBILITY_CHECKS	Audit of why a schedule is eligible, blocked, or requires review.
REMINDER_EVENTS	Concrete planned/sent/failed reminder occurrence; supports idempotent delivery.
ADHERENCE_RECORDS	Patient or authorized-user response such as taken, skipped, or missed.
NOTIFICATIONS	Delivery envelope for reminders and other workflow notifications.

5. Shared-Domain Database Design

Shared-domain views group tables by business responsibility rather than by a single runtime module. They reveal the controls and lifecycle tables that would be difficult to understand in the unified diagram alone.

5.1 Users and Access Control

The access model uses RBAC with optional multiple roles per user. The five initial business roles are patient, HCP, administrator, technical support, and medical reviewer. Permissions are assigned to roles through ROLE_PERMISSIONS; role grant and revocation should record who performed the action and when.

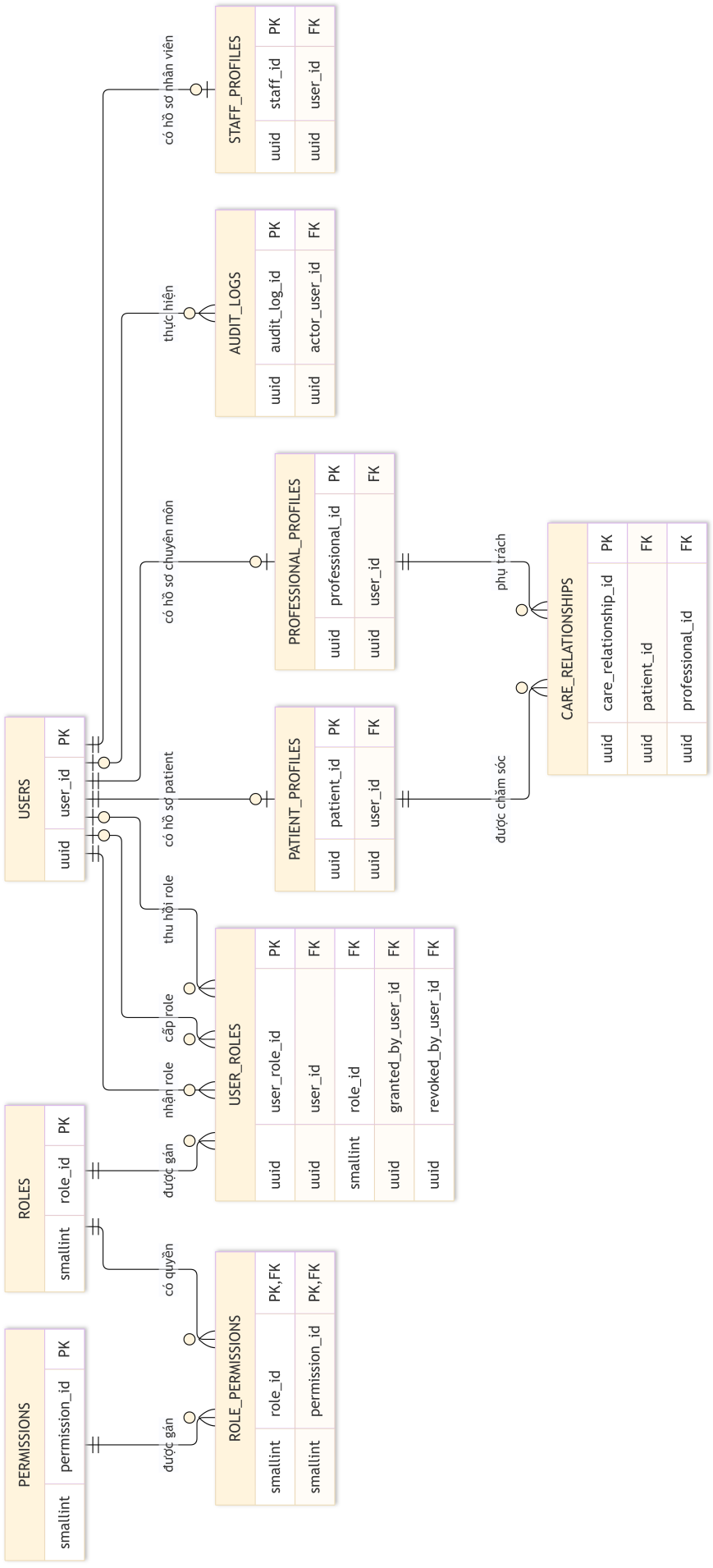

Figure 7: Shared domain – Users and Access Control ERD

Table 9: Users and access-control tables

Table	Responsibility
USERS	Authentication identity, account status, and security metadata.
ROLES	Stable role catalog such as patient, HCP, administrator, technical support, and medical reviewer.
USER_ROLES	Many-to-many role assignment with grant/revoke provenance.
PERMISSIONS	Fine-grained operation catalog used by backend authorization.
ROLE_PERMISSIONS	Role-to-permission mapping; prevents hard-coding every check by role name.
PATIENT_PROFILES	Patient-specific profile and health-context identity.
PROFESSIONAL_PROFILES	Professional qualification, specialty, license, and verification status.
STAFF_PROFILES	Operational employee metadata for administrators, support staff, and reviewers when applicable.
CARE_RELATIONSHIPS	Time-bounded HCP-to-patient authorization relationship.
AUDIT_LOGS	Append-only record of sensitive reads, writes, grants, reviews, and support actions.

The medical reviewer is not a replacement for HCP. A qualified user may hold both roles, but review assignments must be scoped by dataset and task. Technical support may inspect diagnostic information attached to assigned tickets but cannot alter clinical conclusions or approve medication data.

5.2 Medication and Safety Catalog

This domain is the controlled reference catalog. DATASET_VERSIONS groups version-specific rows. Medication products decompose into components and ingredients; safety mappings connect accepted components to safety-source identifiers; interaction pairs connect to participants and evidence.

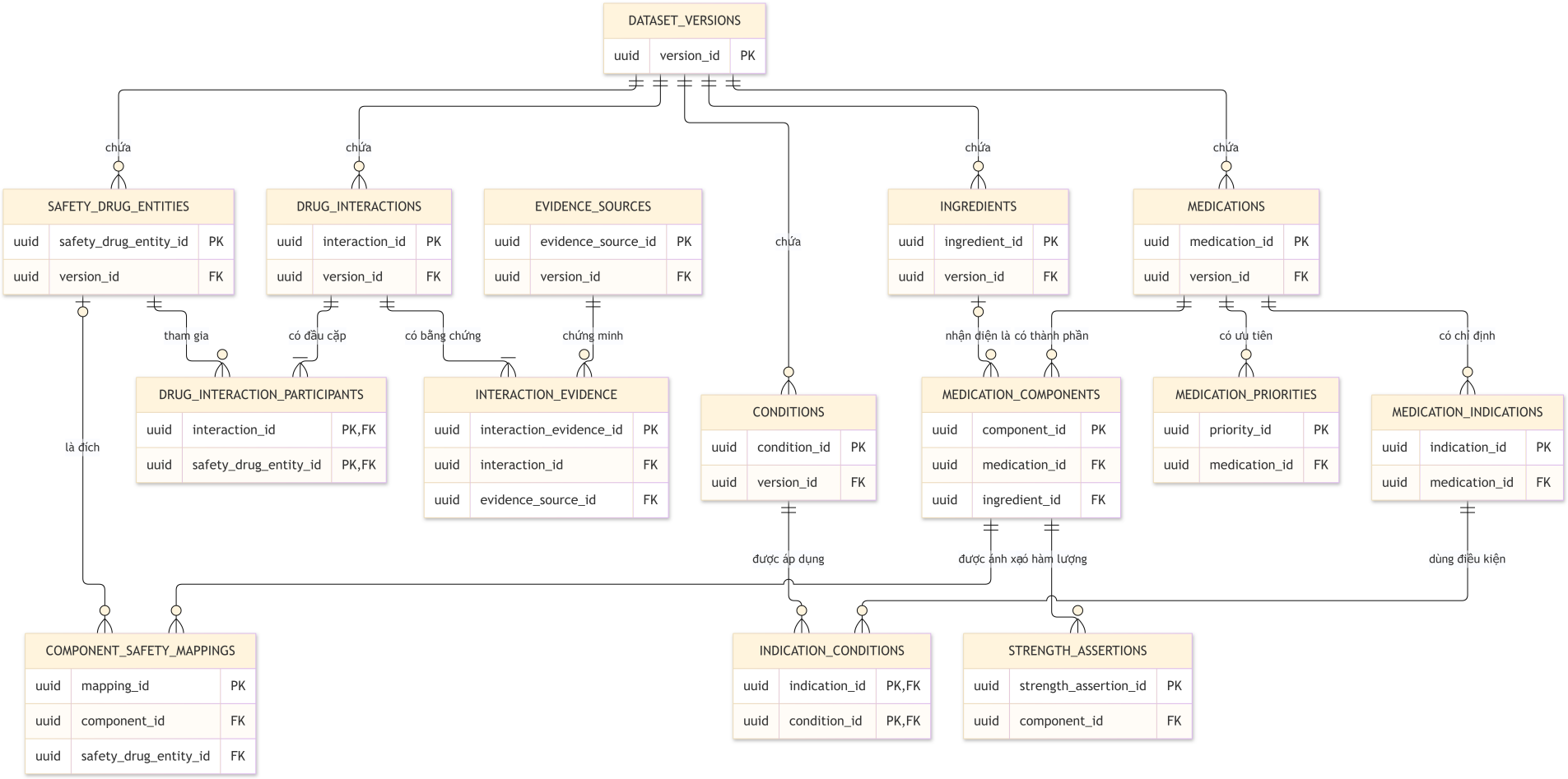

Figure 8: Shared domain – Medication and Safety Catalog ERD

Table 10: Medication and safety catalog tables

Table	Responsibility
DATASET_VERSIONS	Immutable version boundary for imported and reviewed reference data.
MEDICATIONS	Versioned medication product identity.
INGREDIENTS	Versioned ingredient identity and acceptance scope.
MEDICATION_COMPONENTS	Medication composition and optional accepted ingredient link.
STRENGTH_ASSERTIONS	Source and normalized strength statements for a component.
MEDICATION_INDICATIONS	Candidate or approved indication statement for a medication.
CONDITIONS	Normalized, reviewed disease or condition concept.
INDICATION_CONDITIONS	Many-to-many link from indication statements to conditions.
MEDICATION_PRIORITIES	Data-collection and review priority metadata.
SAFETY_DRUG_ENTITIES	Drug identifiers in a safety namespace.
COMPONENT_SAFETY_MAPPINGS	Candidate/accepted/operative mapping from component to safety identifier.
DRUG_INTERACTIONS	Canonical drug interaction with severity and description.
DRUG_INTERACTION_PARTICIPANTS	Interaction endpoints.
EVIDENCE_SOURCES	Provenance record such as file, URL, hash, source type, and retrieval date.
INTERACTION_EVIDENCE	Evidence statements that support or conflict for one interaction.

5.3 Data Import, Review, and Release

Every source import belongs to a batch and version. Automated validation occurs before human review. The assigned medical reviewer makes explicit decisions on review items; evidence is linked to those items. Only an approved version receives a release record for operational use.

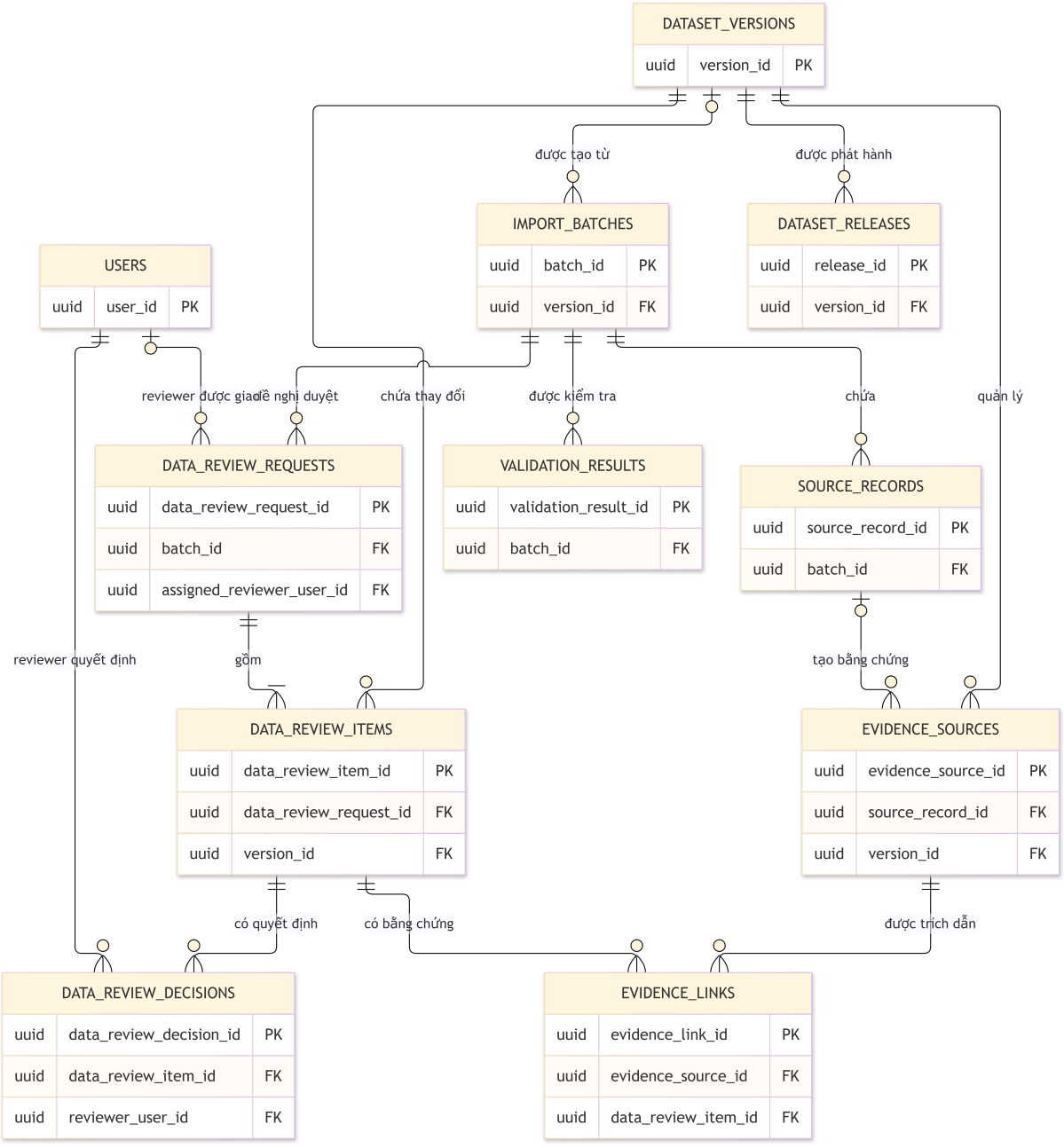

Figure 9: Shared domain – Data Import, Review, and Release ERD

Table 11: Data import and review tables

Table	Responsibility
IMPORT_BATCHES	One ingestion attempt, source package, checksum, importer, and processing state.
SOURCE_RECORDS	Row-level source payload or locator retained for replay and evidence.
VALIDATION_RESULTS	Machine validation output with rule, severity, location, and result.
DATA_REVIEW_REQUESTS	Work package assigned to a medical reviewer.
DATA_REVIEW_ITEMS	Individual entity, field, mapping, conflict, or change requiring review.
DATA_REVIEW_DECISIONS	Append-only reviewer outcome, rationale, scope, and timestamp.
EVIDENCE_LINKS	Links evidence sources to the exact review items they support.
DATASET_RELEASES	Publication record that makes one approved version available to runtime modules.

No separate APPROVAL_DOCUMENTS table is used. Organizational authorization to perform review is represented by user-role history and assignment; the substantive medical decision is represented by DATA_REVIEW_DECISIONS with evidence and audit records.

5.4 Patient Medication Care

This domain connects the patient profile, care relationship, prescription entries, frozen medication list, safety assessment, HCP review, regimen, reminder, and adherence workflows.

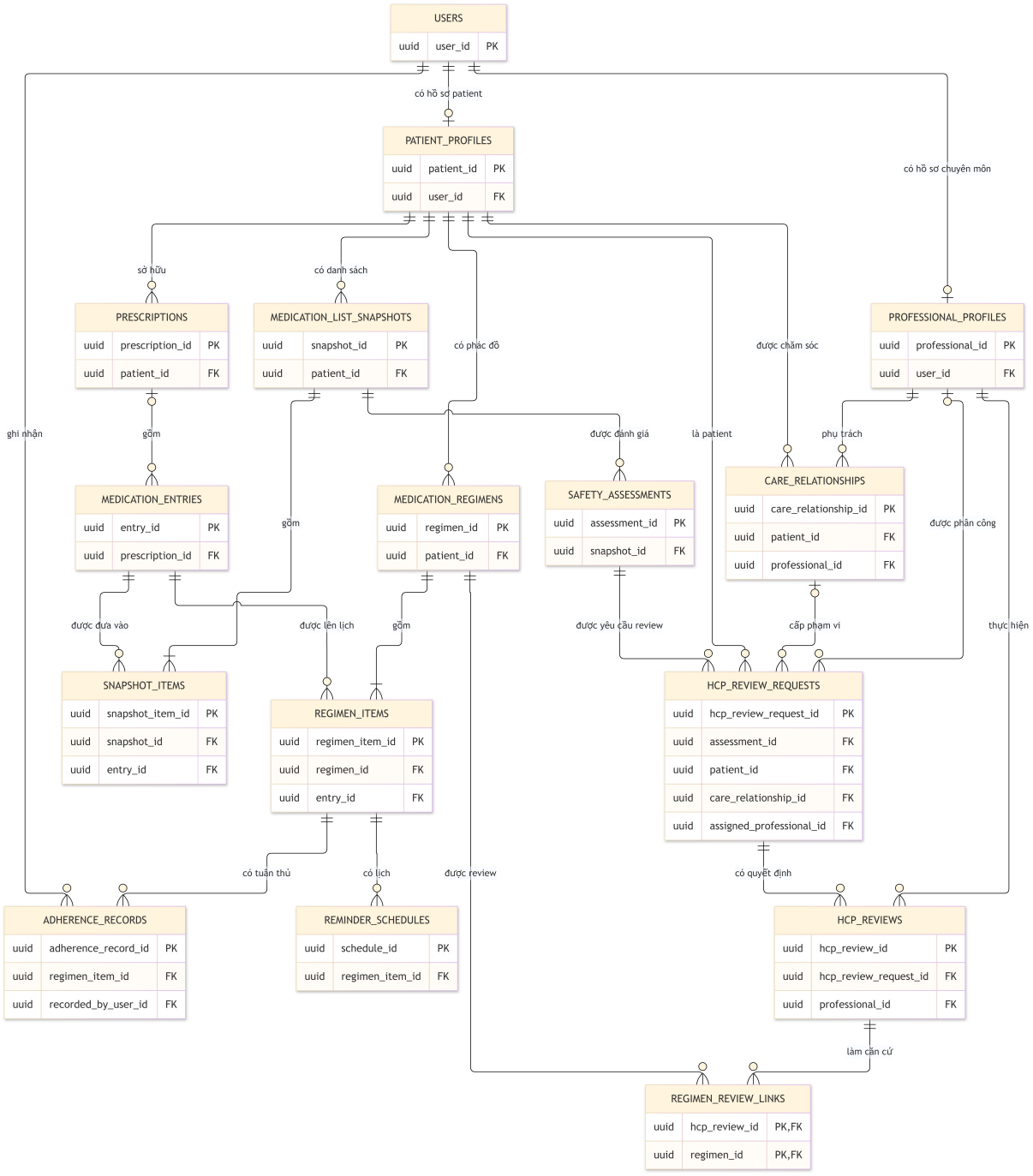

Figure 10: Shared domain – Patient Medication Care ERD

The key boundary is CARE_RELATIONSHIPS: a professional role does not imply access to every patient. A review request references the relationship or equivalent assignment that grants scope. Regimens are connected to the entries they operationalize and to HCP reviews where professional approval is required.

5.5 Communication and Technical Support

The communication model separates professional care from application support. A patient opens a technical support ticket for login, upload, OCR, or failed processing. The ticket is assigned to a support user and receives support-specific messages. Clinical questions and safety interpretation remain in the HCP review conversation.

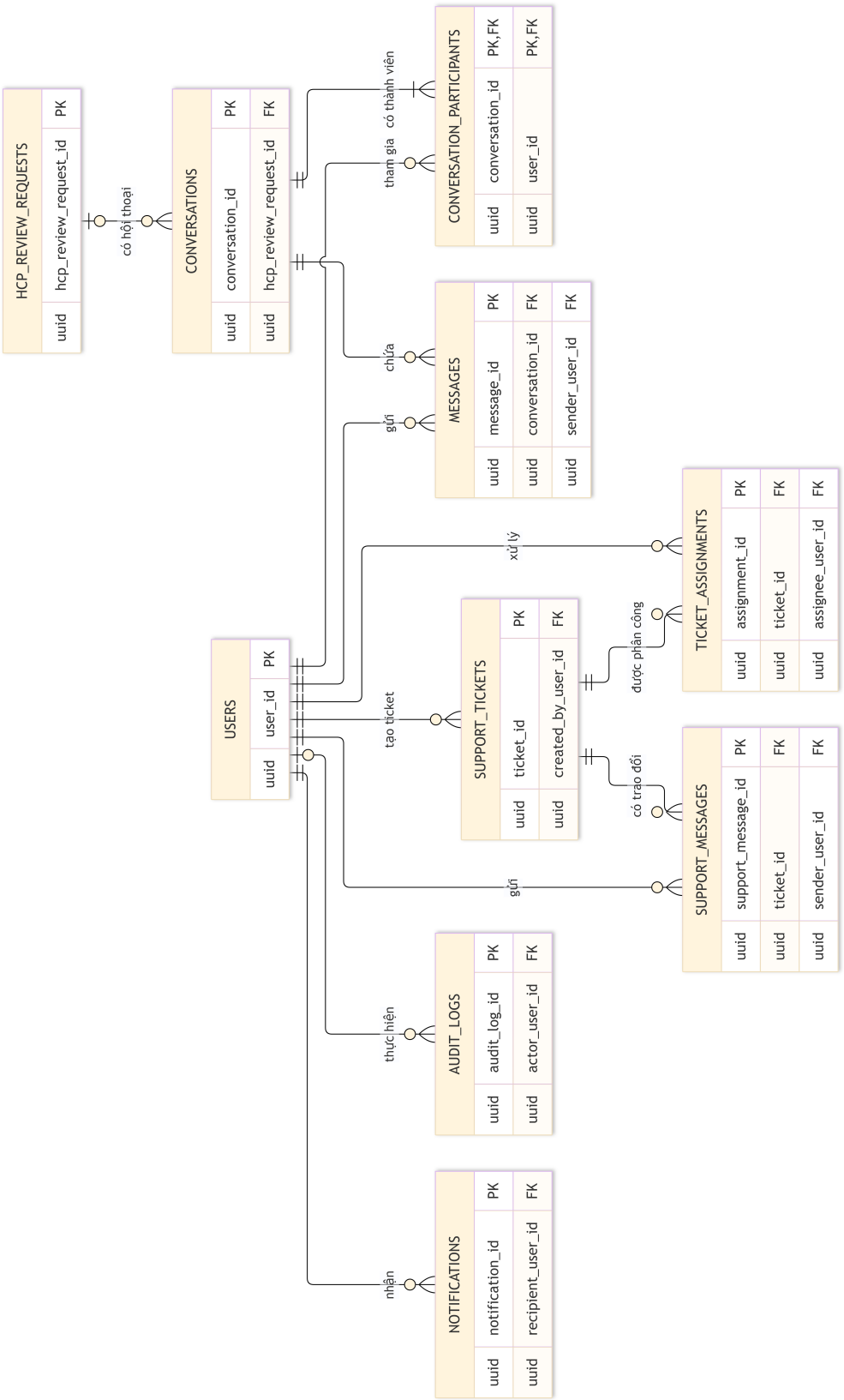

Figure 11: Shared domain – Communication and Technical Support ERD

Table 12: Communication and technical-support tables

Table	Responsibility
CONVERSATIONS	HCP-review-bound clinical communication channel.
CONVERSATION_ PARTICIPANTS	Explicit membership of a clinical conversation.
MESSAGES	Direct patient–HCP messages.
SUPPORT_TICKETS	User-reported technical/operational incident with category, priority, and status.
TICKET_ASSIGNMENTS	Assignment and reassignment history for accountable support ownership.
SUPPORT_MESSAGES	Ticket-specific user/support communication.
NOTIFICATIONS	Recipient-targeted workflow and delivery notifications.
AUDIT_LOGS	Sensitive access and action trace, including support access to diagnostic context.

6. Data Ingestion, Review, and Runtime Flow

6.1 Reference-Data Ingestion Flow

The recommended lifecycle is:

1. Create a draft DATASET_VERSIONS row and one IMPORT_BATCHES row for the source package.
2. Persist source-row identity, relation name, checksum, and raw payload/locator in SOURCE_RECORDS.
3. Transform rows into version-scoped candidate entities while preserving source IDs such as PC_*, LI_*, and CC_* in dedicated source-identifier columns rather than forcing them into UUID primary keys.
4. Execute structural, referential, semantic, and duplicate checks; write all outcomes to VALIDATION_RESULTS.
5. Create DATA_REVIEW_REQUESTS and DATA_REVIEW_ITEMS for unresolved identities, indication normalization, conflicting evidence, and safety mappings.
6. A qualified medical reviewer records accept, reject, request-change, or defer decisions in DATA_REVIEW_ DECISIONS. Each decision includes rationale and evidence links.
7. Mark accepted mappings operative only after all policy conditions are satisfied. Rejected and unresolved candidates remain traceable but cannot be used by runtime assessment.
8. Publish an approved version through DATASET_RELEASES. Runtime services select a release, not an arbitrary draft version.

6.2 CSV-to-Table Mapping Rules

Table 13: CSV ingestion rules

Source	Target	Critical rule
medications.csv	MEDICATIONS	Preserve registration source value, normalized token, route support, issues, review flags, and verification state. Do not infer unsupported route or release characteristics.

Source	Target	Critical rule
ingredients.csv	INGREDIENTS	Preserve source-local identity separately from accepted normalized identity; "international mapping not required" is not proof of international mapping.
medication_ ingredients.csv	MEDICATION_COMPONENTS	Link the accepted ingredient only when verified; keep unresolved components with raw text and review state.
medication_ ingredients.csv	STRENGTH_ASSERTIONS	Preserve the raw expression and parser status. Do not manufacture amount/unit fields from ambiguous text.
medication_ indications.csv	MEDICATION_INDICATIONS	Import as candidate/pending review. Create CONDITIONS and INDICATION_CONDITIONS only after medical normalization.
medication_ priorities.csv	MEDICATION_PRIORITIES	Name columns so collection priority cannot be confused with clinical recommendation.
ddinter_mappings.csv	SAFETY_DRUG_ENTITIES	Create or reuse candidate DDInter identifiers with namespace and catalog hash.
ddinter_mappings.csv	COMPONENT_SAFETY_ MAPPINGS	Import all as non-operative candidates; acceptance requires medical review and evidence.
Separate DDInter pair file	DRUG_INTERACTIONS, DRUG_INTERACTION_ PARTICIPANTS, INTERACTION_EVIDENCE	Canonicalize unordered pairs, create two distinct participants, retain every source claim, and review conflicting severity.

6.3 Patient Assessment Flow

1. Intake and extraction produce medication entries; patient confirmation preserves corrections.
2. Identification uses a released catalog and records all candidates, the selected result, and confidence.
3. A medication-list snapshot freezes the entries and accepted identifications used for assessment.
4. Components resolve to accepted, operative safety entities. Pair generation creates canonical item pairs.
5. Deterministic joins locate reviewed interaction pairs and evidence in the same dataset release.
6. Pair results and findings are persisted. A finding references both the interaction evidence and the affected snapshot items.
7. If interpretation is required, an HCP review request is opened within the patient's care relationship. The patient communicates directly with the assigned HCP.
8. A reviewed regimen may generate schedules and reminder events after eligibility checks.

6.4 Dataset Version and Release Semantics

DATASET_VERSIONS is not the same as the per-row transformation rule_version. The former identifies a coherent database snapshot; the latter identifies the rule that produced or transformed one row. A version may be draft, validating, under review, approved, rejected, or superseded. A release points to an approved version and records publication/retirement metadata. Existing assessments retain their release reference so results are auditable after catalog updates.

6.5 Review Responsibility

The administrator manages accounts, role grants, and operational configuration but does not substitute for medical review. Medical reviewers should normally be qualified pharmacists, clinical pharmacists, physicians for relevant clinical interpretation, or terminology/informatics specialists within a defined scope. Qualification is stored in the professional profile; authorization is granted and audited separately. The assigned reviewer cannot approve outside the assigned data scope, and high-risk changes may require a second-review rule in the physical workflow.

7. Integrity, Security, and Performance Requirements

7.1 Relational Integrity

The physical schema should implement the following invariants in foreign keys, unique constraints, checks, exclusion constraints, or transactions:

- a source identifier is unique within its source namespace and dataset version;
- only approved dataset versions may be released, and at most one active release exists for an intended runtime channel;
- a component may have a null accepted ingredient while awaiting review, but an operative safety mapping must reference accepted entities and an approved decision;
- an interaction contains exactly two distinct safety participants and is unique as a canonical unordered pair within a dataset version;
- an assessment pair uses two distinct snapshot items and is unique after canonical ordering;
- an HCP review author must match the assigned professional or an explicitly recorded reassignment;
- conversation membership is required before reading or sending a message;
- a reminder event is unique for schedule and planned occurrence time, preventing duplicate delivery during retries; and
- audit and review-decision records are append-only from application workflows.

7.2 State Transitions

Status columns must use controlled values and validated transitions. Important state machines include intake, extraction run, medication entry confirmation, identification, dataset version, review request/item, mapping, assessment, HCP review request, regimen, schedule, reminder event, and support ticket. A status alone is insufficient for audit: the actor, timestamp, reason, and prior state should be retained in history or event records for sensitive transitions.

7.3 Access-Control Matrix

Table 14: High-level role access boundaries

Role	May manage	May view	Must not do
Patient	Own intake, confirmations, regimen responses, conversations, and tickets	Own profile, results, HCP responses, reminders	Access another patient's data or approve clinical/reference data.
HCP	Assigned review cases, professional responses, reviewed regimens	Patients covered by active care relationships/assignments	Browse all patients or approve reference data without reviewer authority.
Administrator	Accounts, roles, configuration, release operations subject to policy	Operational metadata and audit needed for administration	Create medical conclusions merely because of administrative status.

Role	May manage	May view	Must not do
Technical support	Assigned tickets, diagnostic notes, retry requests permitted by procedure	Minimum diagnostic context for assigned incidents	Read unrelated health records, alter HCP decisions, or approve drug data.
Medical reviewer	Assigned reference-data review items and decisions	Source evidence and catalog context within review scope	Access patient care records unless separately authorized as HCP.

Authorization must be enforced in the backend and database-access layer; hiding UI controls is not sufficient. Sensitive reads should create audit events where required by policy.

7.4 Privacy and Security Controls

- Store document binaries outside relational tables in encrypted object storage; retain object key, checksum, media type, size, and access policy in `INTAKE_DOCUMENTS`.
- Encrypt data in transit and at rest; manage secrets outside the database and source repository.
- Minimize protected health information in logs, support messages, notifications, and analytics exports.
- Use soft deactivation or policy-controlled retention for accounts and clinical records; do not cascade-delete audit, review, or evidence history.
- Apply malware scanning, file-type validation, and size limits before OCR processing.
- Record dataset hashes and source locators to detect silent source replacement.

7.5 Indexing Recommendations

Table 15: Initial indexing strategy

Access pattern	Recommended index
User role/permission lookup	Composite indexes on `USER_ROLES(user_id,role_id)` and `ROLE_PERMISSIONS(role_id,permission_id)`; unique on each pair.
Patient timelines	Index patient foreign key plus descending creation time on intakes, prescriptions, snapshots, assessments, regimens, and tickets.
Medication identification	Version/release plus normalized registration token, normalized name key, and accepted ingredient key.
Interaction lookup	Unique canonical pair key within version/release; indexes on participant safety entity and interaction ID.
Review work queues	Partial/composite indexes on assignee, status, priority, and creation time.
Reminder dispatch	Index active schedule and next due time; unique event key on schedule and scheduled time.
Audit investigation	Actor/time and target-type/target-ID/time indexes; consider time-based partitioning when volume requires it.
Support queue	Assignee/status/priority and creator/created-time indexes.

Indexes should be validated against actual query plans after representative data is loaded. Large text, raw JSON, and evidence payloads should not be indiscriminately indexed.

7.6 Transactions and Concurrency

Critical operations require transactions: accepting a review decision and changing mapping status; publishing a release; freezing a medication snapshot; writing assessment results; assigning review/support work; and generating idempotent reminder events. Optimistic version columns are recommended for mutable workflow rows. Long-running OCR and assessment jobs should update job state through controlled transitions rather than hold database transactions open.

7.7 Backup, Retention, and Observability

Backups must include the relational database and coordinated object-storage versions. Restore tests should verify that document references and hashes still resolve. Operational metrics should cover import failures, validation severity, review backlog, assessment latency, unresolved identification rate, reminder delivery failure, and support SLA without exposing medication or message content in metric labels.

8. Design Decisions and Next Steps

8.1 Vector Embedding Decision

Vector embedding is **not required** for the authoritative drug–drug interaction check. The safety path should use reviewed ingredient identity, accepted component-to-safety mappings, canonical interaction pairs, and relational joins. A similarity score cannot establish that two drug identities are equivalent and must not independently generate or suppress a medical warning.

Embeddings may be added later as an auxiliary retrieval feature for:

- proposing medication or ingredient candidates from noisy OCR text;
- retrieving similar unresolved names for reviewer triage; or
- searching evidence passages semantically.

If introduced, embeddings should be stored separately with model name, model version, source-text hash, creation time, and entity reference. Candidate retrieval must still pass deterministic validation and, for safety mappings, medical review. This preserves a clean boundary between probabilistic assistance and operative clinical data.

8.2 Recommended Physical Platform

PostgreSQL is a suitable implementation target because the design benefits from UUIDs, robust constraints, transactional integrity, JSON support for preserved source payloads, partial indexes, and optional row-level security. This is a recommendation rather than a committed project constraint. The physical design should standardize UUID generation, timestamp/timezone handling, controlled status values, naming conventions, and migration tooling before schema creation.

8.3 Open Design Decisions

Table 16: Open decisions before physical schema approval

Decision	Options/impact	Required owner
Database engine and deployment topology	PostgreSQL version, managed/self-hosted, environments, HA, backup objectives	Technical lead
Patient identity and minimum profile	Required demographics, identity verification, consent, account recovery	Product owner and privacy reviewer
Professional/reviewer qualification	License fields, verification process, review scopes, second-review thresholds	Clinical owner and administrator

Decision	Options/impact	Required owner
Dataset release policy	Approval quorum, rollback, supersession, active channels, effective dates	Medical data owner
Interaction severity vocabulary	Source-specific levels versus normalized severity and conflict handling	Clinical reviewer
Retention and deletion	Intake documents, messages, tickets, audit, evidence, and health records	Privacy/legal owner
Notification provider and delivery policy	Channels, retry, quiet hours, timezone, escalation, idempotency	Product and technical leads
Physical column dictionary	Types, nullability, defaults, checks, unique constraints, and indexes for all 64 tables	Database designer

8.4 Implementation Sequence

1. Approve entity boundaries, role responsibilities, review workflow, and release semantics in this logical model.
2. Produce a physical data dictionary for all tables, including types, nullability, constraints, and state values.
3. Implement access control, dataset governance, and reference catalog first because later modules depend on them.
4. Build a repeatable CSV staging/import pipeline with validation reports and reviewer work queues.
5. Load a non-operative draft dataset and verify row counts, unresolved records, hashes, and source traceability.
6. Add the separate interaction-pair dataset and evidence before enabling safety assessment.
7. Implement patient workflows incrementally: intake, identification, snapshot/assessment, HCP review, then regimen/reminder.
8. Run integrity, authorization, migration, backup/restore, performance, and clinical-review acceptance tests before any production release.

8.5 Conclusion

The proposed model separates patient operations from governed reference data while preserving the joins needed for end-to-end medication safety. Its most important control is the boundary between candidate data and released, operative data: OCR output, indication text, and DDInter mappings remain reviewable evidence until accepted by an authorized reviewer. With the remaining physical decisions resolved, the 64-entity logical model can be converted into a normalized SQL schema without changing the core workflows documented in PA2.